SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No. : 3

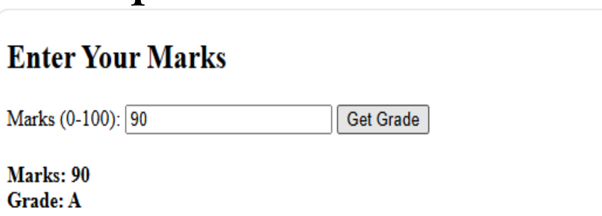
1. Title: Implement control structures and form validation; create a grading system based on user-entered marks. .

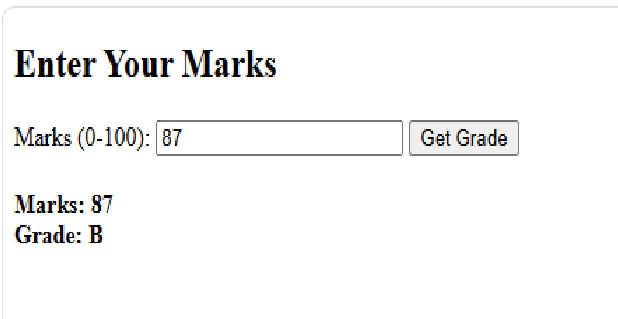
2. Software/Tools Required: VS code,Web browser

3. Theory:

This experiment demonstrates the use of **control structures** and **form validation** in JavaScript by creating a grading system based on user-entered marks. Form validation ensures that the entered marks are valid (not empty, numeric, and within the range of 0–100). Control structures such as **if-else** or **switch** are then used to evaluate the marks and assign the appropriate grade. The result is displayed dynamically, making the application interactive, accurate, and user-friendly.

4. Output:





Code:

```
<!DOCTYPE html>
<html>
<head>
```

```
    <title>Grading System</title>
    <script>
        function validateAndGrade() {
            // Get the input value
            var marksInput =
document.getElementById("marks").value.trim();
            var marks = parseFloat(marksInput);
            var result = document.getElementById("result");

            // Clear previous result
            result.innerHTML = "";

            // Validate input: check if input is a number
            if (marksInput === "" || isNaN(marks)) {
                alert("Please enter a valid number for marks.");
                return;
            }

            // Validate range: marks should be between 0 and 100
            if (marks < 0 || marks > 100) {
                alert("Marks should be between 0 and 100.");
                return;
            }

            // Determine grade based on marks using control
structures
            var grade;
            if (marks >= 90) {
                grade = "A";
            } else if (marks >= 80) {
                grade = "B";
            } else if (marks >= 70) {
                grade = "C";
            } else if (marks >= 60) {
                grade = "D";
            } else {
                grade = "F";
            }

            // Display the result
            result.innerHTML = "Marks: " + marks + "<br>Grade: " +
grade;
        }
```

```
    </script>
</head>
<body>
    <h2>Enter Your Marks</h2>
    <form onsubmit="event.preventDefault();
validateAndGrade();">
        <label for="marks">Marks (0-100):</label>
        <input type="text" id="marks" name="marks" />
        <button type="submit">Get Grade</button>
    </form>
    <div id="result" style="margin-top:20px; font-
weight:bold;"></div>
</body>
</html>
```

5.Result/Conclusion

The grading system was successfully developed using JavaScript to validate user-entered marks and assign grades accurately. The experiment demonstrated the effective use of **control structures** (if-else/switch) and **form validation** to ensure valid input and correct grade calculation. It enhanced the understanding of decision-making, input validation, and dynamic result display, helping in the development of interactive and reliable web applications.